

Cross-Level Business-Metric Diagnosis: Association & $Y|X$ Shift

Two complementary, kernel/causal-objective tools — no explicit metric formulas required

Business metrics live at different levels (order, user, merchant) with no explicit relationships. Two questions, two tools:

(A) Which metrics move together? — HSIC (association linkage). For metric pairs (a, b) with RBF Gram matrices K, L and centering $H = I - \frac{1}{n}\mathbf{1}\mathbf{1}^\top$,

$$\widehat{\text{HSIC}}(a, b) = \frac{1}{(n-1)^2} \text{tr}(KHLH) \geq 0, \quad = 0 \iff a \perp\!\!\!\perp b.$$

Symmetric, non-linear, assumption-light; large $\widehat{\text{HSIC}}$ = a linkage edge. It recovers the chain and isolates unrelated metrics (*no direction claimed*).

(B) Did a target's $Y|X$ change, and on which X ? — PO-risk (one model per target). With batch W as “treatment”, cross-fit $\mu(x) = \mathbb{E}[Y|x]$, $e(x) = \Pr(W=1|x)$; pseudo-outcome $\psi = (Y - \mu)(W - e)$; $\tau = \arg \min_f \mathbb{E}[(\psi - f(X))^2]$;

$$\text{PO-risk} = \mathbb{E}[\tau(X)^2] (> 0 \Rightarrow P(Y|X) \text{ shifted}), \quad \phi_j = \text{PO-risk}(X) - \text{PO-risk}(X_{-j}).$$

One meta-learner takes all X jointly; LOCO ϕ_j localizes the change — cost $O(\#\text{targets})$, not $O(p^2)$ pairwise models.

(A) HSIC linkage			(B) PO-risk $Y=\text{GMV}$ (test $p = 0.010$)	
a	b	$\widehat{\text{HSIC}}$	metric x_j	LOCO ϕ_j
delivery	rating	0.055	rating (driver)	+0.106
delivery	gmv	0.046	orders	+0.046
rating	n_orders	0.062	delivery	+0.039
n_orders	gmv	0.070	promo	+0.032
* w/ quantity/discount		≤ 0.002	tenure	+0.028
Chain <code>delivery--rating--n_orders--gmv</code> recovered; noise metrics isolated.			$P(Y X)$ shifted; change loads on rating .	

Use together. HSIC maps the metric linkage across levels; for any target metric, PO-risk says whether its $Y|X$ relationship shifted and which linked metric carries it — so a change in a merchant-level metric can be localized to an order-level metric, without modelling every pair and without a causal-proportion claim.